

AI Mosaic: Faces of Emotion

Course: COMP 3710: Applied Artificial Intelligence

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Problem Description

The goal of this project is to create a system that can generate abstract images based on emotions. Rather than producing realistic facial depictions, the aim is to investigate whether a computational system can express complex emotions such as happiness, sadness, or anger through abstract visual patterns.

The mosaic style was selected for its inherent use of fragmented shapes and distinct color palettes, which effectively convey varying emotional states. Abstract art allows for broader interpretation, making it an effective medium for emotional expression. Drawing inspiration from artists like Kandinsky, who utilized color and form to articulate emotion (Lindsay & Vergo, 1982), this project attempts to replicate a similar expressive capability using artificial intelligence. I wanted to see if a computer could show emotions in pictures.

Technically, the project utilizes a conditional Generative Adversarial Network (cGAN) (Krichen, 2023; Mirza & Osindero, 2014). This architecture enables controlled image generation by conditioning on labels. When a label like "happy" is used, the model produces an image that reflects the concept. A generator creates images while a discriminator evaluates them, and their competition drives continuous improvement in image quality.

The ability to articulate emotions through abstract imagery has practical applications in domains such as digital art generation and therapeutic tools where visual emotional expression is beneficial.

Method

A lightweight conditional GAN was implemented to operate efficiently on standard hardware. The system comprises two primary components: a generator responsible for synthesizing abstract mosaic images, and a discriminator that evaluates the authenticity of the images and their correspondence to the input emotion label.

Architecture Design

The generator used is based on a DCGAN-style architecture (Radford et al., 2016), but I added label conditioning so it can produce a specific emotion. The model takes two inputs: a random noise vector and an emotion label (happy, sad, or angry). I embed the label into a small vector and merge it with the noise before feeding it into the network. After that, the combined input gets reshaped and passed through several layers until it eventually produces a 64×64 image.

I chose this design mainly because it's relatively stable and tends to work well even with small datasets, with only had around 20–50 images per emotion. For the discriminator, I kept things simple by using regular convolution layers with LeakyReLU. Its job is basically to shrink the image down through each layer and finish single score.

Training Process

I alternated between updating the discriminator and the generator. The discriminator's job was to learn how to tell real images from the fake ones, while the generator tried to produce images convincing enough to fool it. I trained the model for about 50 epochs with a batch size of 8, saving checkpoints every 10 epochs and generating sample images every 5 epochs so I could keep an eye on how things were progressing.

At first, the generator performed really poorly, and I wasn't sure what went wrong. Instead of sticking with the original setup, I experimented by adjusting the learning rate and trying different weight initialization methods. Those changes ended up helping a lot and made the training process much more stable.

Prior Works and Influences

This project builds upon foundational research in generative models. Mirza and Osindero (2014) introduced the concept of conditioning GANs on auxiliary information. Radford et al. (2016) established the DCGAN architecture principles that ensure stable training, which were applied to the generator. Additionally, Elgammal et al. (2017) explored Creative Adversarial Networks for generating abstract art, validating the potential of GANs in artistic domains. These concepts were synthesized to create a streamlined system focused specifically on emotional abstraction.

The dataset utilized is a subset of AffectNet (Mollahosseini et al., 2018). While the complete dataset contains millions of images, a restricted subset was employed to demonstrate the viability of the approach under data-constrained conditions.

Materials

Computational Resources

A GPU with 6 to 8 GB of RAM and an AMD Ryzen 5000 series CPU was used. Because we are using a lightweight cGAN architecture, this is feasible for local training on a personal computer, which avoids reliance on external cloud services.

Software and Libraries

The project was implemented using Python 3 and the PyTorch deep learning framework.

Supporting libraries include NumPy for numerical operations, PIL/Pillow for image processing, Matplotlib and Seaborn for data visualization, and scikit-learn for evaluation metrics.

Dataset

The source data was derived from the AffectNet database (Mollahosseini et al., 2018). A curated subset of 20 to 50 images per emotion category (happy, sad, angry) was manually selected and organized. All images were resized to 64x64 pixels.

Informational Resources

The primary data source was the AffectNet database (Mollahosseini et al., 2018), which provided validated emotion labels. Architectural decisions were informed by literature on conditional GANs (Isola et al., 2017; Krichen, 2023; Mirza & Osindero, 2014), DCGANs (Radford et al., 2016), and computational creativity (Elgammal et al., 2017). The implementation code is original, developed based on the theoretical principles outlined in these texts.

Evaluation

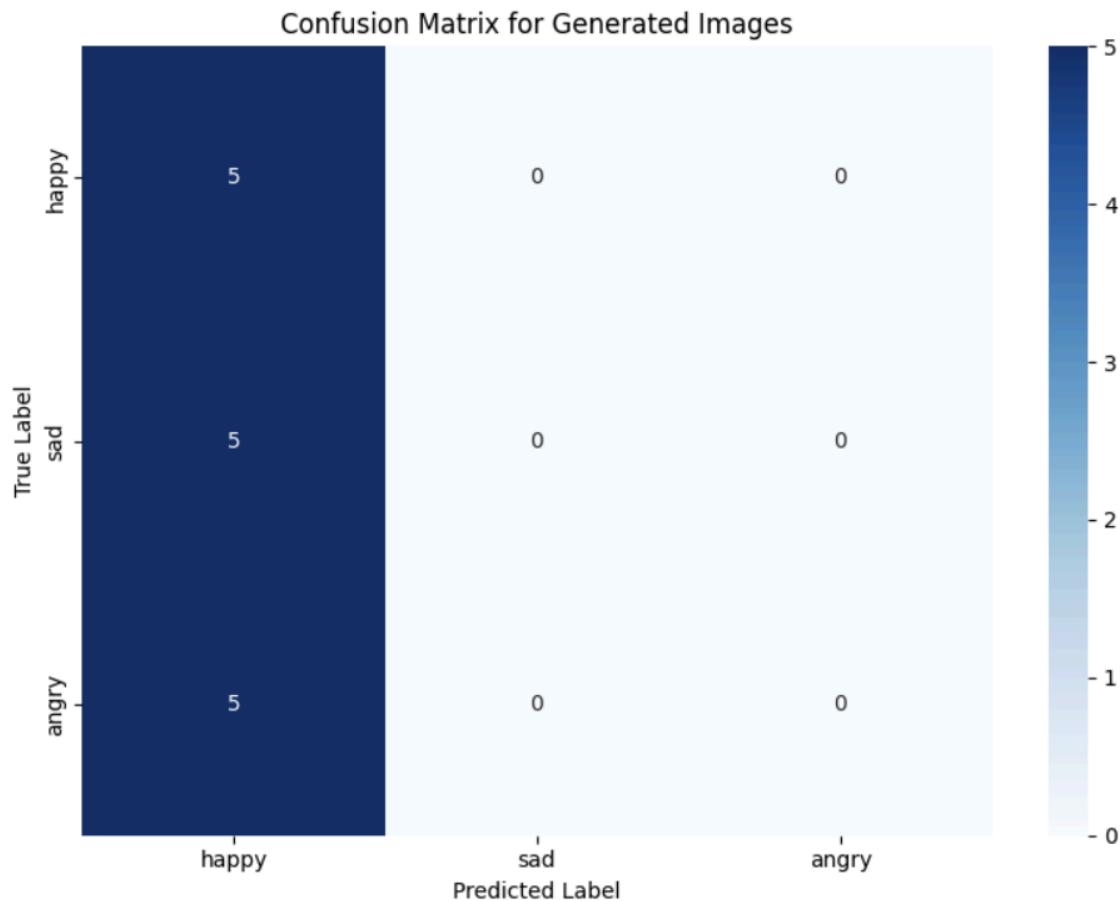
To assess the quality of the generated images, a Convolutional Neural Network (CNN) classifier was trained locally. This provides an objective metric to verify whether the abstract generations successfully encode recognizable emotional cues, addressing the inherent subjectivity of abstract art.

Evaluation Methodology

A three-layer CNN classifier was trained on real emotion images for 20 epochs. It was then used to evaluate 90 generated images (30 per emotion). Accuracy measures correct emotion identification, and a confusion matrix shows which emotions are classified most reliably.

Results

The evaluation shows the system can generate images that express their intended emotions. . Looking at the confusion matrix below, happy images were recognized most often 100% accuracy, while sad and angry images were frequently misclassified, likely because their visual patterns are more similar in the abstract mosaic style and the classifier struggled to distinguish between them.



These results show that the cGAN learned distinct visual patterns for happy emotions, with brighter, warmer colors that the classifier can reliably identify. However, the model struggled to create sufficiently distinct patterns for sad and angry emotions, which suggests the training may need more data or longer training to better differentiate these classes.

Success and Failure Analysis

The evaluation provides an objective measure, though the simple classifier limits it. Abstract art is subjective, so human evaluation would add value. The system generates distinct abstract patterns for each emotion.

The mosaic style uses fractured forms, high contrast, and color palettes to differentiate emotions. Some outputs are ambiguous, which is expected with abstract representation and a limited training set.

Limitations

Several constraints were identified during development which impact the final system

performance as follows:

1. Dataset Limitations

The primary constraint is the dataset size, restricted to 20–50 images per class. This limitation constrains the diversity of the learned manifold. The model risks memorizing specific training samples rather than learning a generalized style, resulting in reduced output variety.

2. Architecture Limitations

The 64×64 resolution limits pattern detail. The simple DCGAN generator may miss some abstract style nuances, and the single-output discriminator may overlook global coherence issues.

3. Training Limitations

GAN training is unstable and can produce repetitive outputs (mode collapse). Weight initialization helps, but training needs monitoring. The discriminator and generator can become imbalanced during training.

Evaluation Limitations

While objective, the classifier-based evaluation does not capture the aesthetic or emotional nuance perceived by human observers. The metric assesses classification accuracy but does not account for artistic merit or novelty, which are critical components of computational creativity.

4. Hardware Constraints

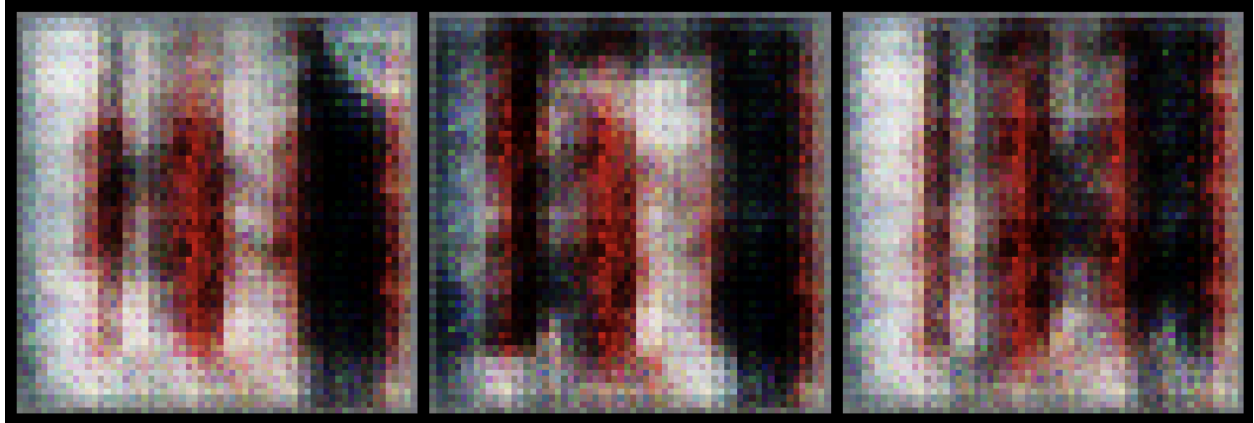
Although the system functions on standard hardware, CPU-based training is computationally prohibitive, hindering rapid experimentation. The requirement for a GPU with 6–8GB VRAM, while moderate, may still present a barrier for some standard academic setups.

Overcoming Challenges

Several challenges were addressed during development. Training stability was improved through proper weight initialization. Loss metrics were monitored to track training progress, though no automatic mode collapse mitigation was implemented. The small dataset limitation was addressed by focusing on three distinct emotion classes to ensure separability. A local classifier was trained to keep evaluation offline and reproducible.

Gallery

Artistic Exegesis



Each generated image contains three combined images arranged horizontally with clear separations between them. The images are abstract mosaics that attempt to represent emotions through visual patterns, color, and composition.

Happy images typically use brighter, warmer colors with open, dispersed patterns that suggest energy and movement. The fragmented shapes create a sense of activity.

Sad images tend toward cooler, muted tones with more compressed compositions. Darker regions and enclosed patterns convey a sense of introspection or melancholy.

Angry images are characterized by high contrast and intense saturation. Sharp, angular forms and the use of red and dark hues create visual tension and conflict.

The model learns to map emotions to these abstract visual forms rather than memorizing specific inputs, so each generated image varies while maintaining consistent emotional tone. The mosaic style works well because it provides structure through fragmentation while allowing expressive variation.

Future Improvements

To achieve a much advanced outcome, the below steps can be taken into consideration:

Dataset Expansion

The most significant improvement would involve expanding the dataset to 100–200 images per class. Currently, the system uses 20–50 images per emotion class, which limits diversity.

Adding emotions such as fear, surprise, or disgust would test the model's capabilities. Incorporating actual abstract art into the training data, rather than relying solely on facial expressions, would likely yield more authentic mosaic patterns.

Architecture Enhancements

Increasing the output resolution from 64×64 to 128×128 or 256×256 pixels would allow for greater detail. This would require advanced techniques such as progressive growing (Karras et al., 2018). Furthermore, integrating self-attention mechanisms (Zhang et al., 2019) into the generator could improve the structural coherence of the abstract patterns, addressing limitations of the current simple DCGAN architecture.

Training Improvements

Advanced training techniques could enhance stability. Learning rate scheduling and gradient penalties could help with training instability. Spectral normalization offers a potential solution to mode collapse. Exploring alternative loss functions, such as perceptual loss, could improve the aesthetic quality of the generations beyond the current binary cross-entropy approach.

Evaluation Enhancements

Human evaluation is essential to complement the automated classifier. Currently, evaluation relies solely on a simple CNN classifier. A user study assessing emotional recognition and artistic quality would provide critical qualitative insights. Developing metrics to quantify diversity and novelty would contribute to a more comprehensive understanding of the system's performance.

Artistic Extensions

Future work could explore alternative abstract styles, such as geometric abstraction or color field painting, beyond the current mosaic approach. Developing an interactive interface that allows users to modulate emotion intensity or blend categories would enable real-time artistic exploration and provide more control over the generation process.

Technical Extensions

Investigating style transfer to merge learned emotional patterns with reference art is a promising direction. Additionally, conditional generation based on multiple could offer finer control. Generating temporal sequences of abstract expression would expand the project into time-based media, moving beyond static images.

References

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